PT-4o illustration for “CycleGAN of life”

CS-229

Greg Ver Steeg

This week:

Finish unsupervised learning

- A few missing pieces on Flow Matching
- GAN
- VAE
- VQ-VAE
- VQ-GAN

Next week (week 9):

- Monday: Memorial day (no class)

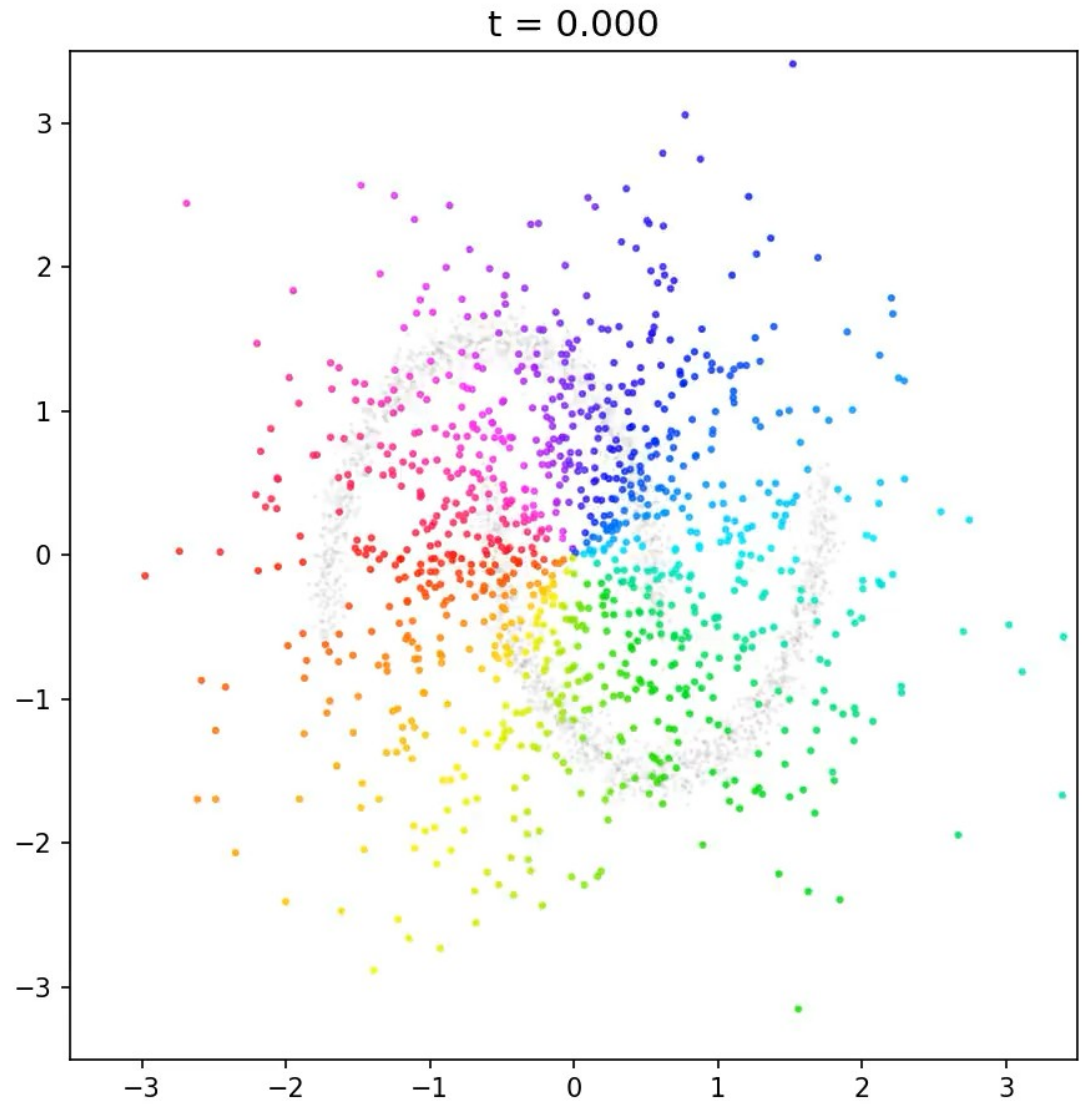
Finish flow matching

With one movie and a healthy dose of math

I missed the best illustration in the demo

Start with z(t=0) from Gaussian (color points to make it easier to track where they go)

Simulate $\dot{z} = \hat{v}(z, t)$



Detail I missed from JIT!

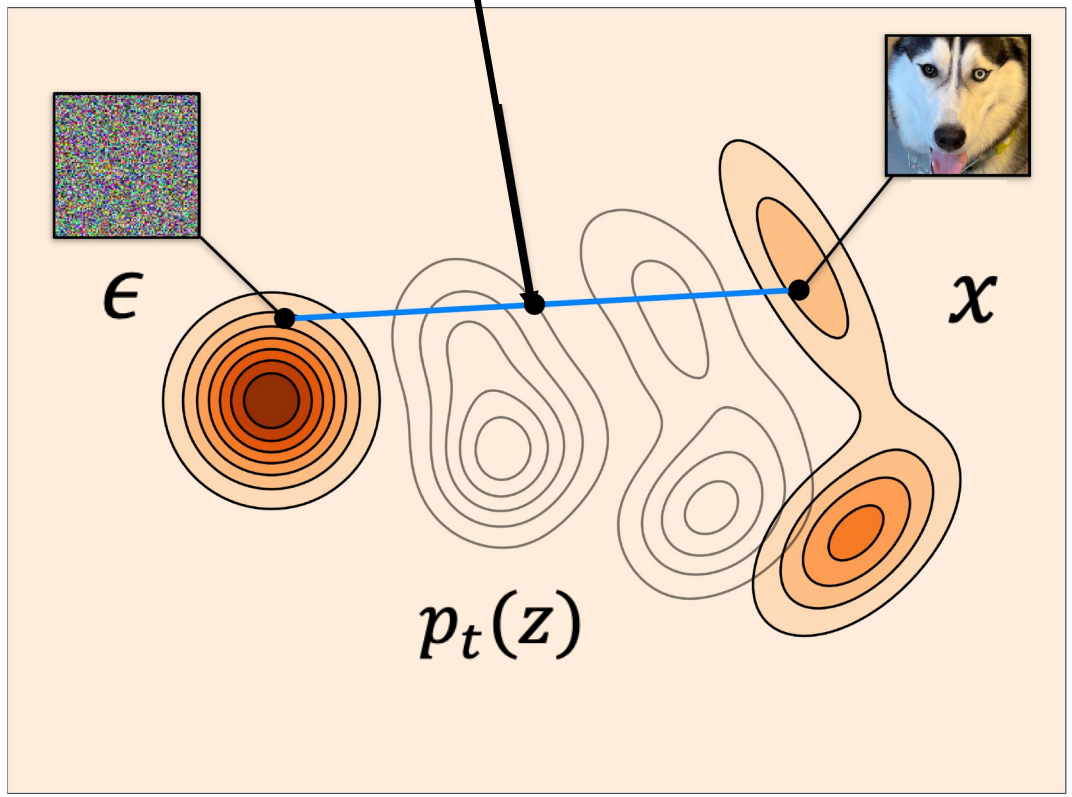
In HW 3 I'll suggest that:

- sample the logit normal during training, but

- constant linear time steps for sampling

	JiT-B	JiT-L	JiT-H	JiT-G
architecture				
depth	12	24	32	40
hidden dim	768	1024	1280	1664
heads	12	16	16	16
image size	256 (other settings: 512, or 1024)			
patch size	`image_size` / 16			
bottleneck	128 (B/L), 256 (H/G)			
dropout	0 (B/L), 0.2 (H/G)			
in-context class tokens	32 (if used)			
in-context start block	4	8	10	10
training				
epochs	200 (ablation), 600			
warmup epochs [17]	5			
optimizer	Adam [31], $\beta_1, \beta_2 = 0.9, 0.95$			
batch size	1024			
learning rate	2e-4			
learning rate schedule	constant			
weight decay	0			
ema decay	{0.9996, 0.9998, 0.9999}			
time sampler	$\text{logit}(t) \sim \mathcal{N}(\mu, \sigma^2)$, $\mu = -0.8$, $\sigma = 0.8$			
noise scale	$1.0 \times$ `image_size` / 256			
clip of $(1 - t)$ in division	0.05			
class token drop (for CFG)	0.1			
sampling				
ODE solver	Heun [20]			
ODE steps	50			
time steps	linear in [0.0, 1.0]			
CFG scale sweep range [22]	[1.0, 4.0]			
CFG interval [33]	[0.1, 1] (if used)			

$$z_t = (1-t)\epsilon + t\,x$$



$$\dot{z}_t = \underbrace{x - \epsilon}_{v}$$

Train a neural network to predict “v”

Let’s show that this works – i.e. using:

$$\dot{z}_t = \hat{v}(z,t)$$

Gives the same distributions!
(*for optimal* $\hat{v}$)

Loss

$$\mathbb{E}_{t,\epsilon,x}\left[\|\hat{v}(z,t) - v\|\text{^}2\right]$$

Derive the "unconditional" flow

$z_t = (1-t)\epsilon + t\,x$

$$\dot{z}_t = v = x - \epsilon = \frac{x - z}{1 - t}$$

Use "Continuity equation" to get dynamics of $p_t(z|x)$

$$\partial_t p_t(z|x) = -\nabla \cdot (p_t(z|x)\, v) = -\nabla \cdot \left(p_t(z|x)\, (x - z)/(1 - t)\right)$$

Take expectation over p(x)

$$\int dx\, p(x) \partial_t p_t(z|x) = \partial_t p_t(z)$$

$$= \int dx\, p(x) [-\nabla \cdot \left(p_t(z|x) \frac{(x - z)}{(1 - t)} \right)]$$

$$= -\nabla \cdot (p_t(z) \frac{(\mathbb{E}[x|z,t] - z)}{(1 - t)})$$

Warning: notation and style of derivation a bit different from FM lite

Conditional mean minimizes mean square error

$$\arg\min_{\hat{x}(z,t)} \mathbb{E}_{p_t(z|x)p(x)}[|x - \hat{x}(z,t)|^2]$$

Then:

$$\hat{x}(z,t) = \mathbb{E}_{p_t(x|z)}[x] = \mathbb{E}[x|z,t]$$

Conclusion: the min MSE denoiser gives the correct “flow”

Tweedie's formula

$$\nabla_z \log p_t(z) = \frac{t}{(1-t)^2}\mathbb{E}[x|z,t] - \frac{z}{(1-t)}$$

The expression on the left is called the "score".

Shows up everywhere.

Score is related to optimal denoiser!!

*Higher order Tweedie identities: Dytso et al "*Meta Derivative Identity for the Conditional Expectation"

Diffusion – reverse SDE dynamics

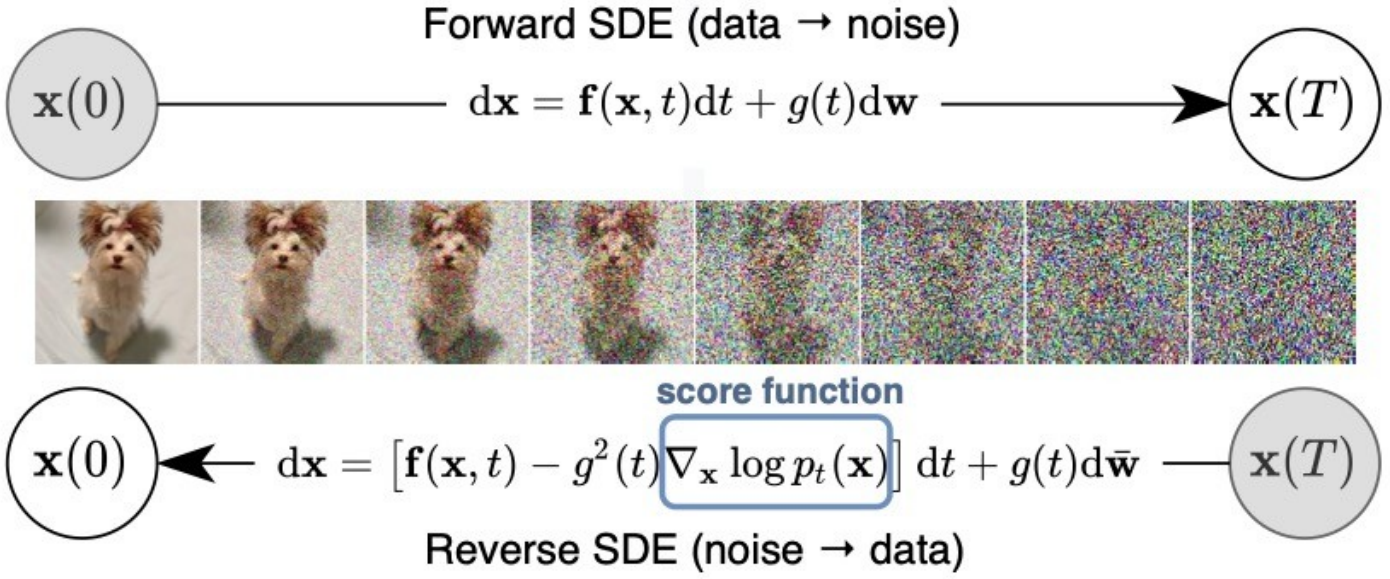


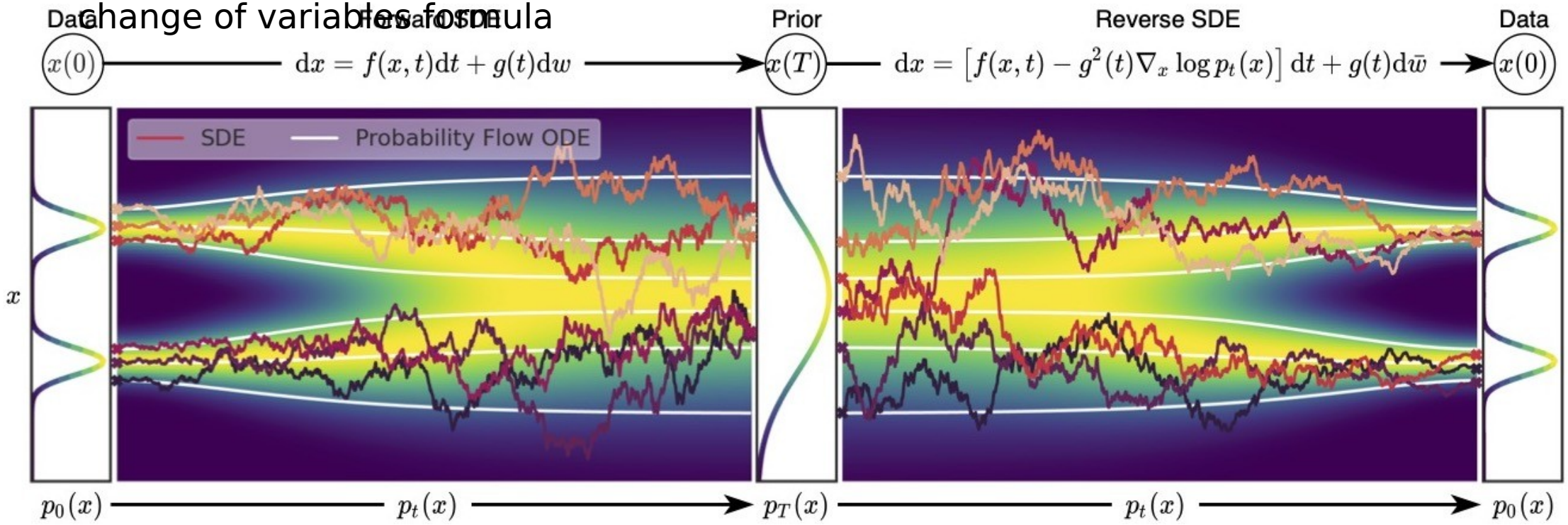
Figure 1: **Solving a reverse-time SDE yields a score-based generative model.** Transforming data to a simple noise distribution can be accomplished with a continuous-time SDE. This SDE can be reversed if we know the score of the distribution at each intermediate time step, $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

Famous paper:
Score-Based Generative Modeling through Stochastic Differential Equations

Convert to continuous flow?

This ODE has the same marginal densities. (Fokker Planck again!)
So if we replace the learned score function, we can solve this ODE to get probability density estimates, with the change of variables formula

$$\mathrm{d}\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right] \mathrm{d}t.$$



Elucidating the Design Space of Diffusion-Based Generative Models

Tero Karras NVIDIA
Miika Aittala NVIDIA
Timo Aila NVIDIA
Samuli Laine NVIDIA

Abstract

We argue that the theory and practice of diffusion-based generative models are currently unnecessarily convoluted and seek to remedy the situation by presenting a design space that clearly separates the concrete design choices. This lets us identify several changes to both the sampling and training processes, as well as preconditioning of the score networks. Together, our improvements yield new state-of-the-art FID of 1.79 for CIFAR-10 in a class-conditional setting and 1.97 in an unconditional setting, with much faster sampling (35 network evaluations per image) than prior designs. To further demonstrate their modular nature, we show that our design changes dramatically improve both the efficiency and quality obtainable with pre-trained score networks from previous work, including improving the FID of a previously trained ImageNet-64 model from 2.07 to near-SOTA 1.55, and after re-training with our proposed improvements to a new SOTA of 1.36.

Continuous normalizing flow – just needs the score function

$$\mathrm{d}\boldsymbol{x} = -\dot{\sigma}(t)\,\sigma(t)\,\nabla_{\boldsymbol{x}} \log p\big(\boldsymbol{x}; \sigma(t)\big)\,\mathrm{d}t,$$

Then Fokker Planck equation says that
p(x; sigma(t)) evolves from Gaussian to target distribution under these dynamics!

- How do we get $\nabla_{\boldsymbol{x}} \log p\big(\boldsymbol{x}; \sigma(t)\big)$

- It happens to be MSE!

$$\nabla_x \log p(x,t) \propto -\hat{\epsilon}(x,t)$$

Where the denoiser optimizes:

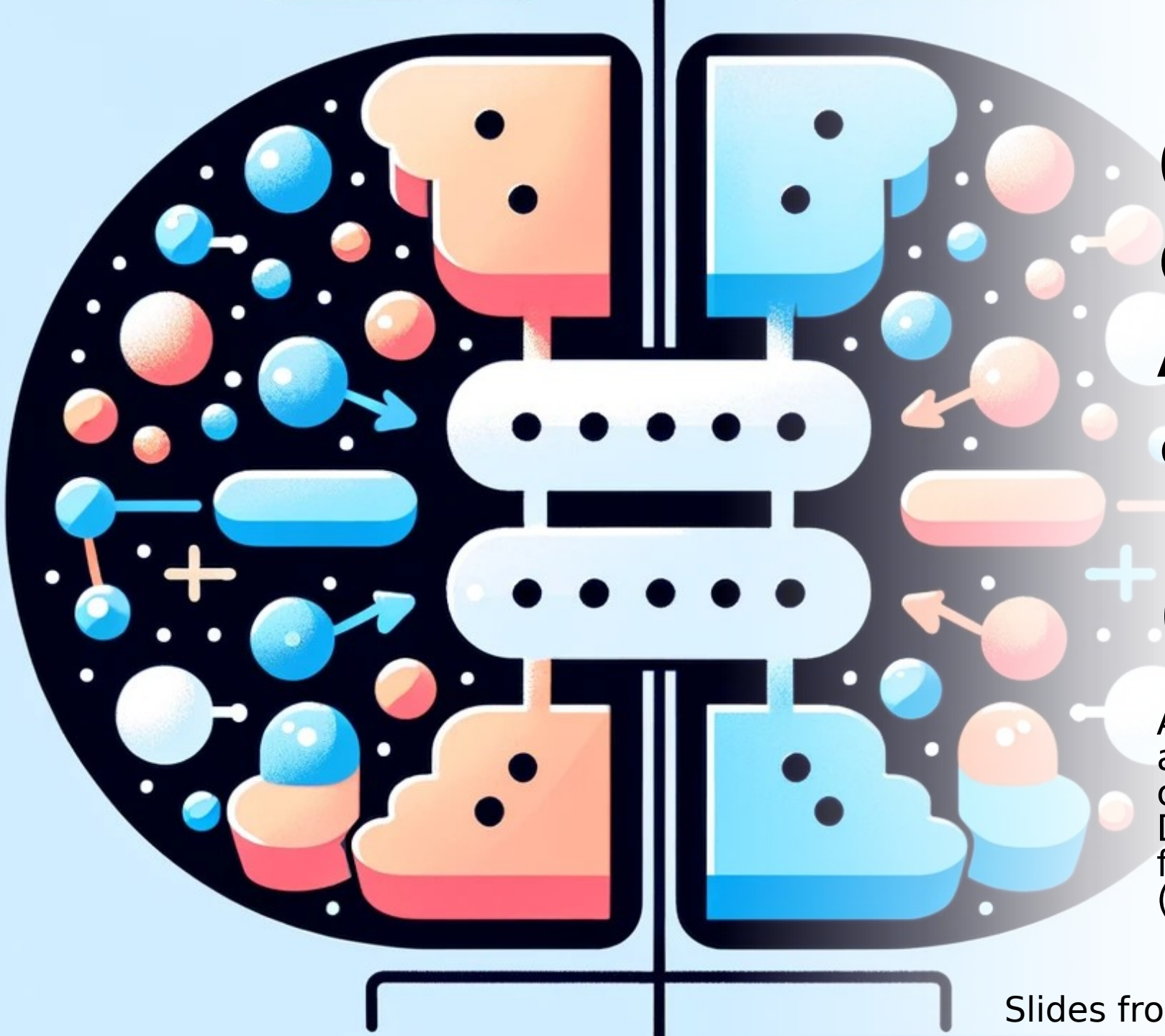
$$\min_{\hat{\epsilon}} E[|\epsilon - \hat{\epsilon}(x,t)|^2]$$

The same kind of MSE objective we get from VAE and flow matching perspective! (but maybe weighted differently)

Continuous normalizing flow – just needs the score function

$$\mathrm{d}\boldsymbol{x} = -\dot{\sigma}(t)\,\sigma(t)\,\nabla_{\boldsymbol{x}} \log p\big(\boldsymbol{x}; \sigma(t)\big)\,\mathrm{d}t,$$

Then Fokker Planck equation says that
p(x; sigma(t)) evolves from Gaussian to target distribution under these dynamics!

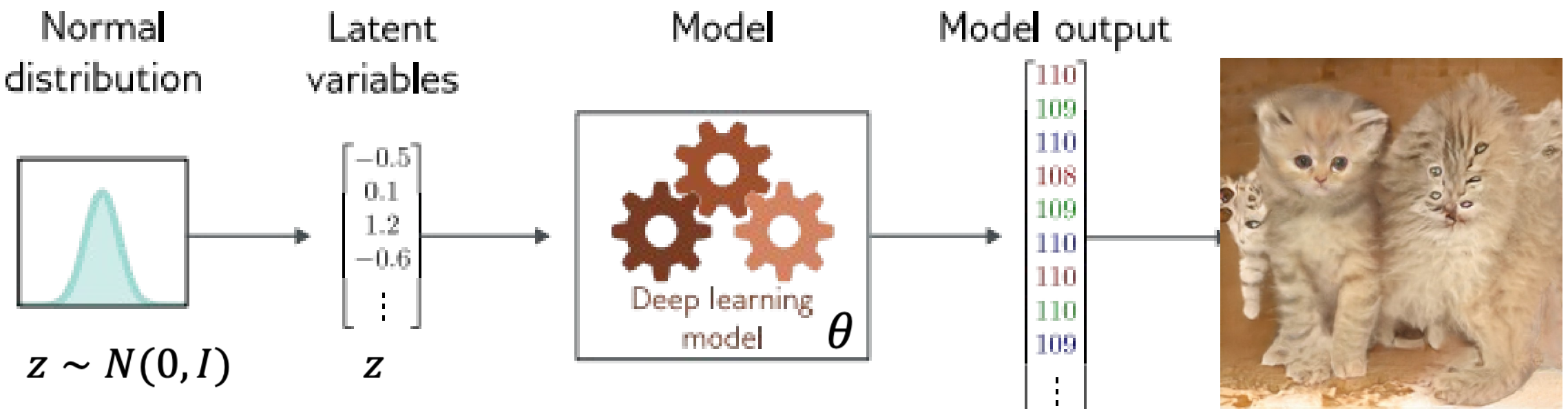


Generative Adversarial Networks (GANs)

A few unique ideas in ML, and new work tries to combine GANs with Diffusion, or uses ideas from GANs in new context (like CycleGAN)

Slides from https://udlbook.github.io/udlbook/

Latent variable models (VAE or GAN)

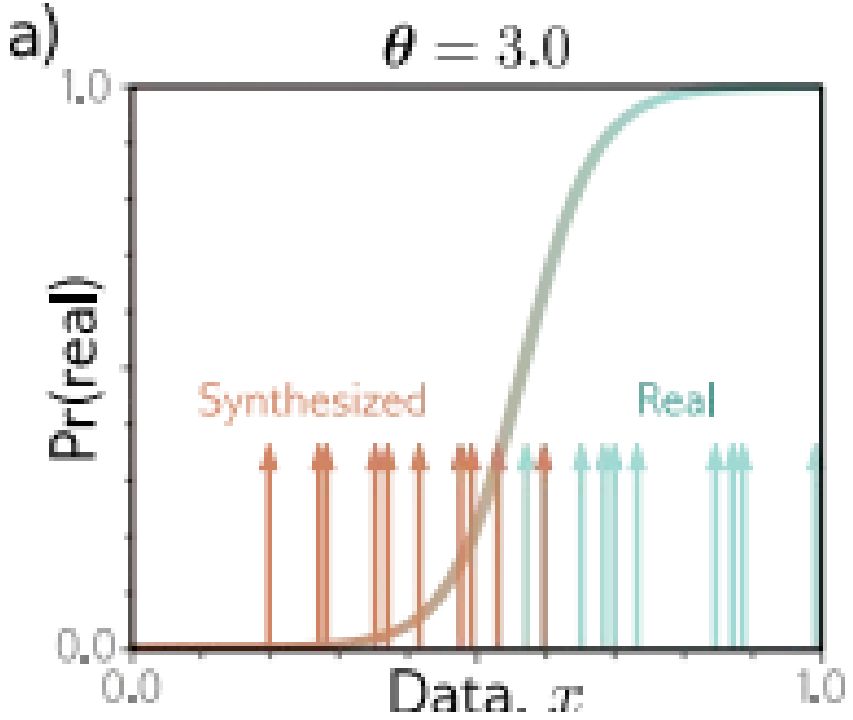


Latent variable models map a random “latent” variable to create a new data sample
In GANs, this is called the generator:

$$\mathbf{x}_* = \mathbf{g}[\mathbf{z}_i, \boldsymbol{\theta}]$$

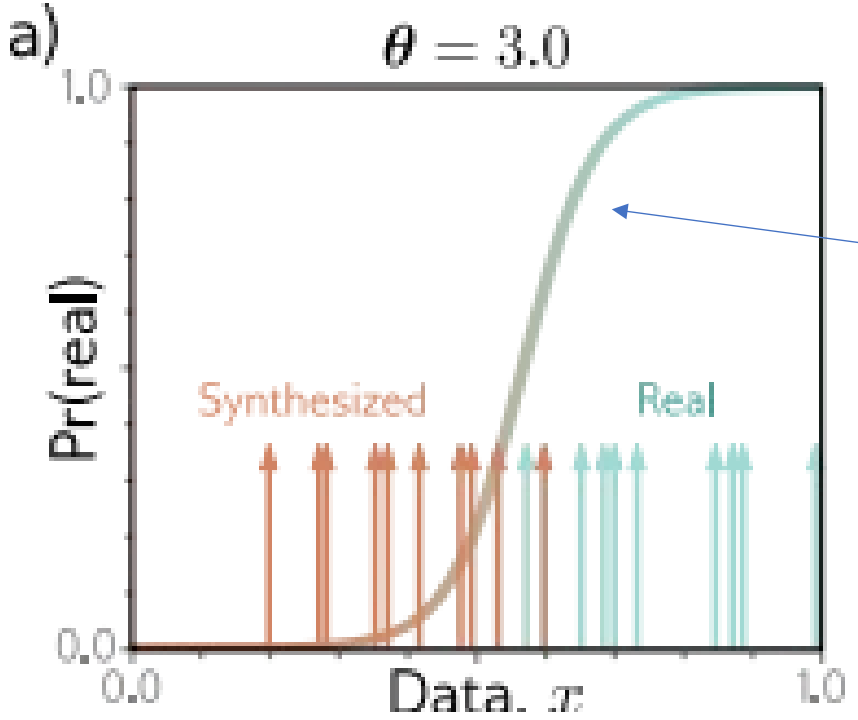
GAN example

$$x_j^* = \mathrm{g}[z_j, \theta] = z_j + \theta$$



GAN example

$$x_j^* = \mathrm{g}[z_j, \theta] = z_j + \theta$$

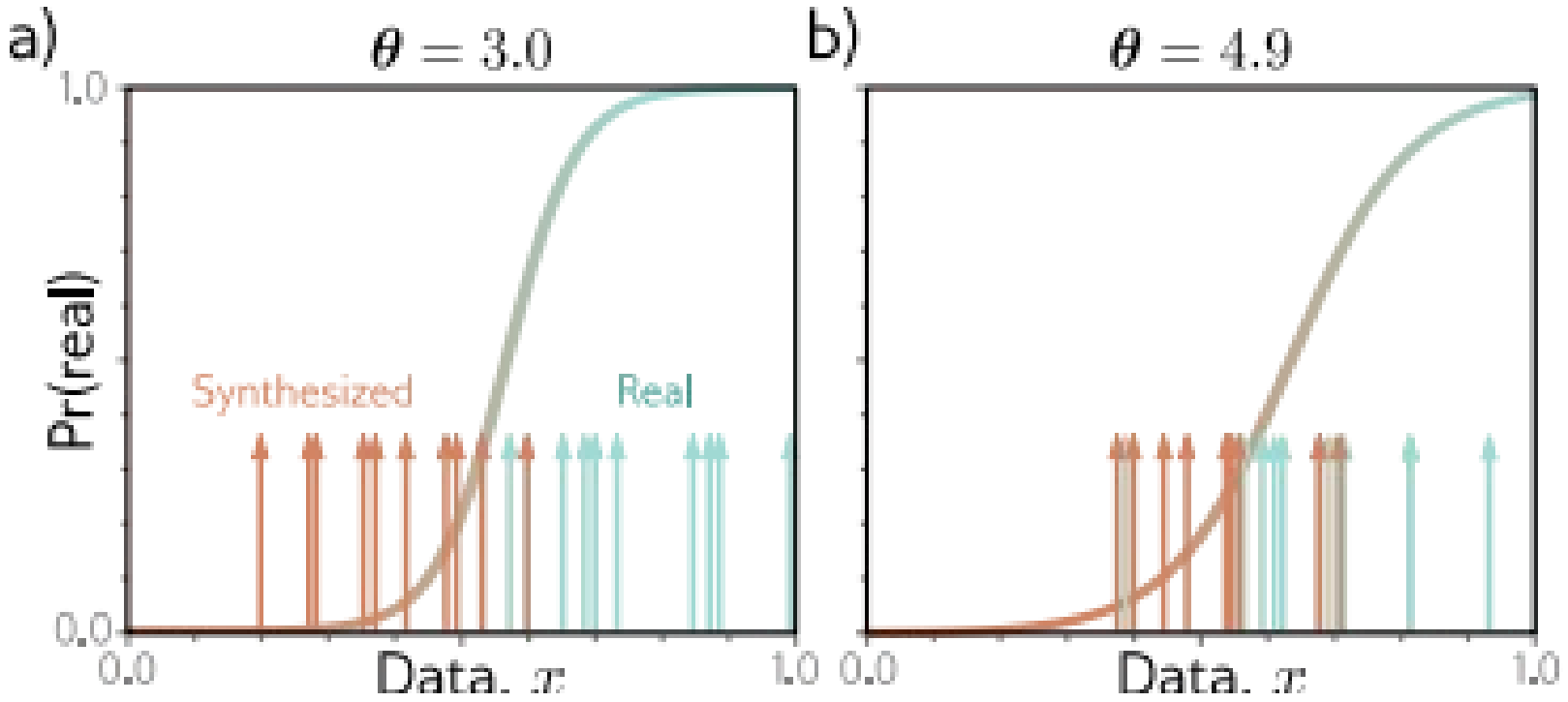


$f[\bullet, \phi]$

Discriminator

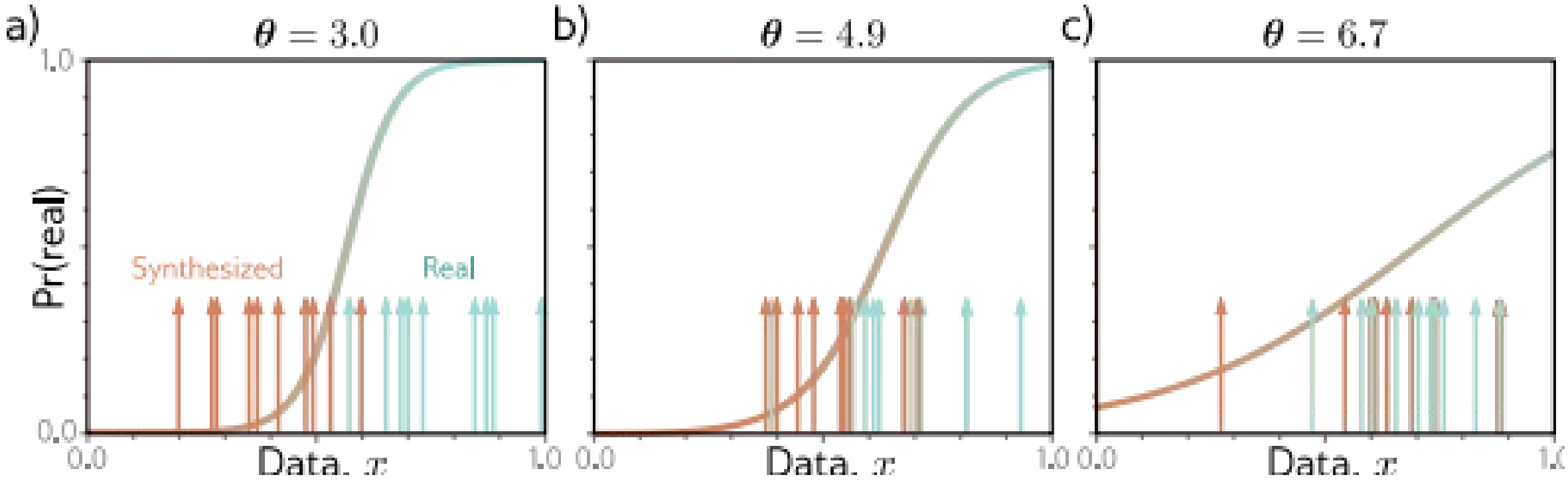
GAN example

$$x_j^* = \mathrm{g}[z_j, \theta] = z_j + \theta$$



GAN example

$$x_j^* = \mathrm{g}[z_j, \theta] = z_j + \theta$$

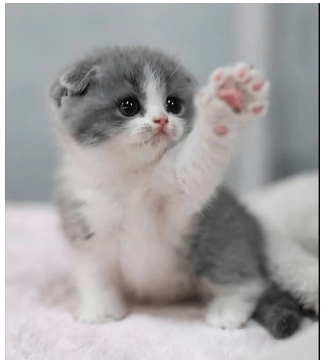


GAN cost function

Discriminator uses standard cross entropy loss:

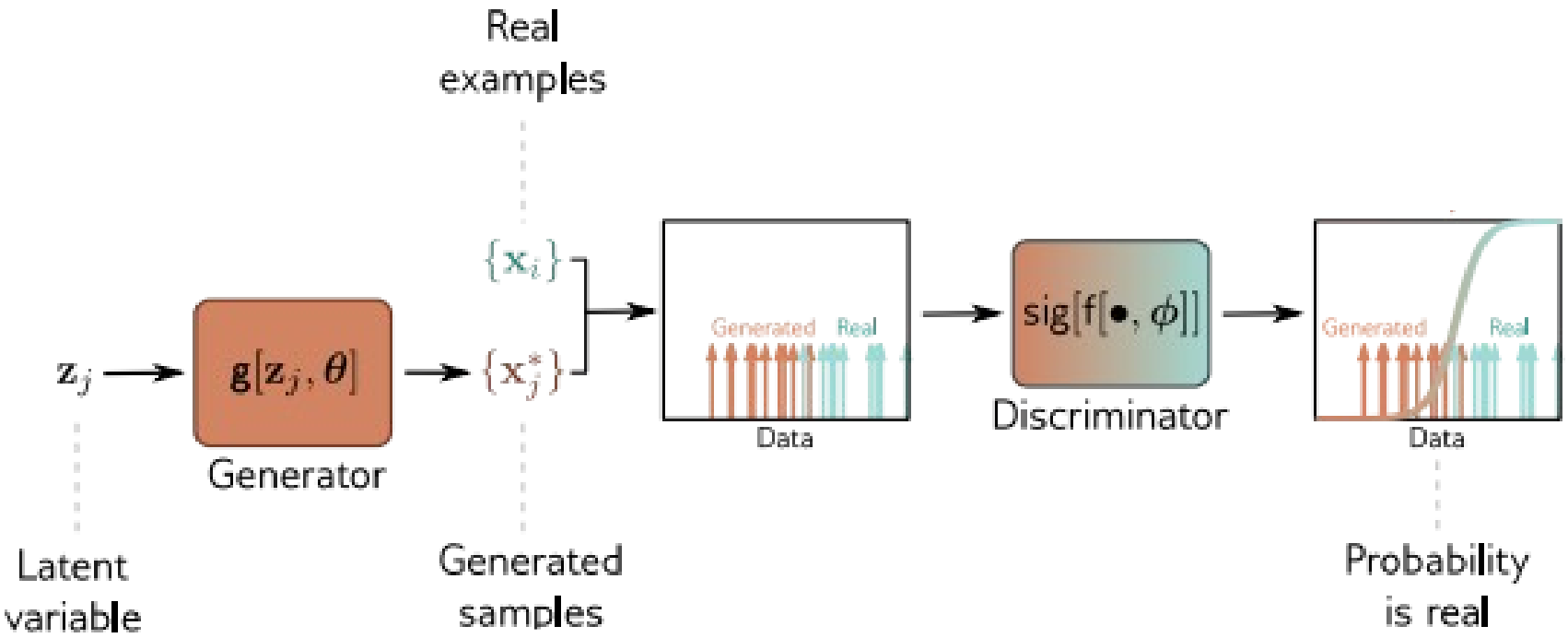
$$\hat{\phi} = \underset{\phi}{\operatorname{argmin}} \left[\sum_i -(1-y_i) \log\Big[1 - \operatorname{sig}[\mathrm{f}[\mathbf{x}_i, \phi]]\Big] - y_i \log\Big[\operatorname{sig}[\mathrm{f}[\mathbf{x}_i, \phi]]\Big] \right]$$

x_i , $y_i = 0$ FAKE

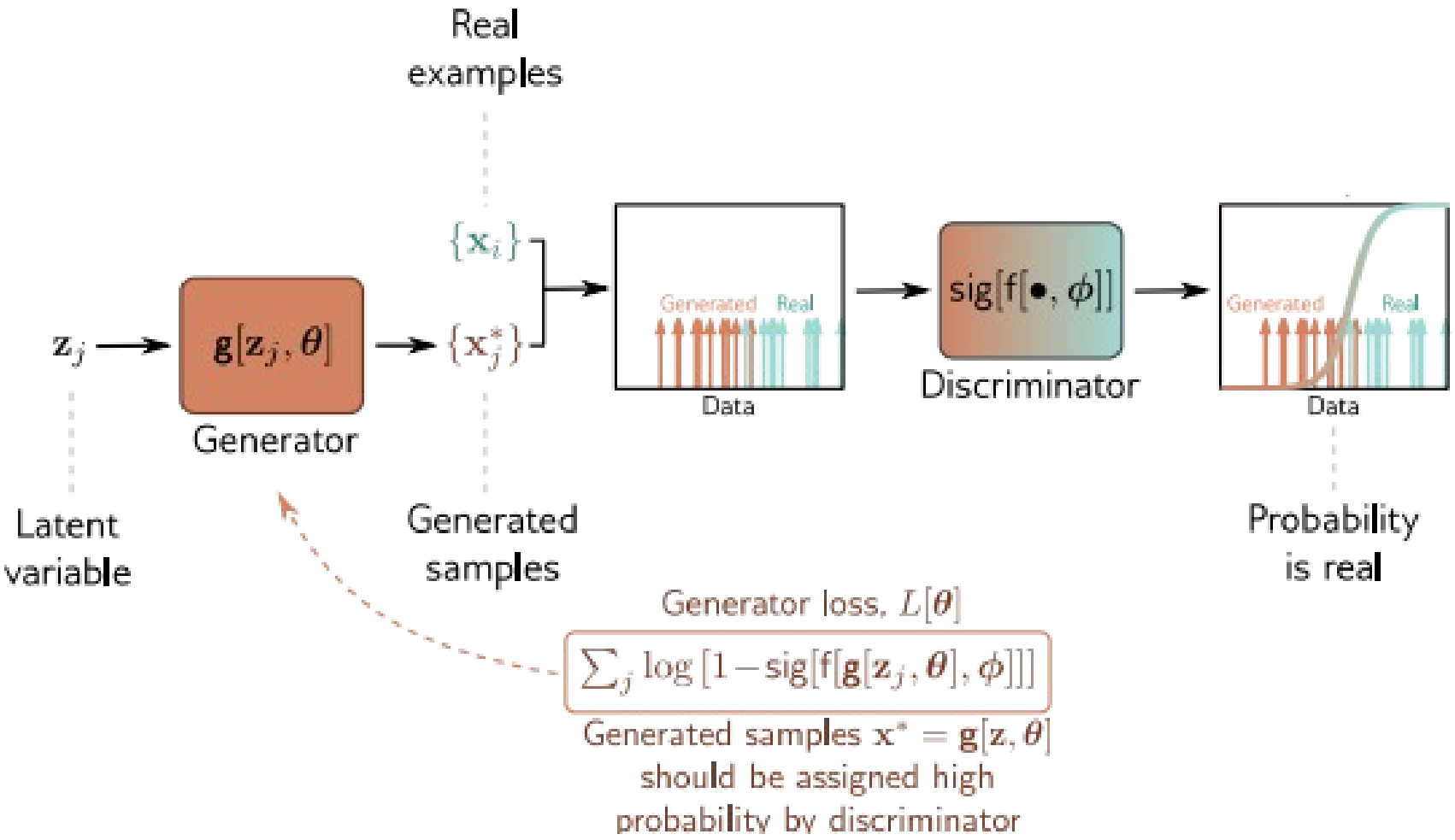


x_i , $y_i = 1$ REAL

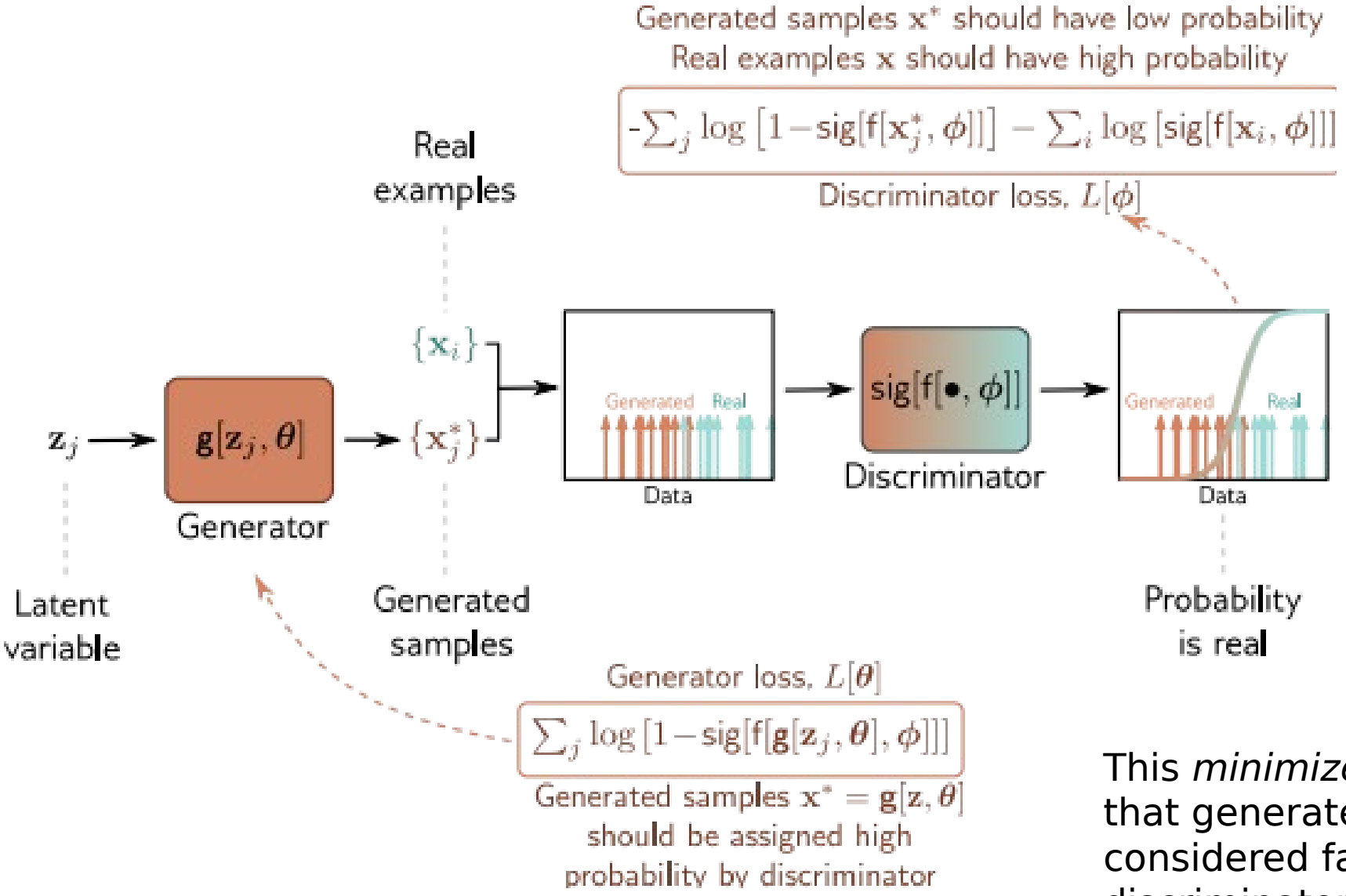
GAN loss function



GAN loss function



GAN loss function



This *minimizes* probability that generated image is considered fake, by discriminator

GAN cost function

Discriminator uses standard cross entropy loss:

$$\hat{\boldsymbol{\phi}} = \underset{\boldsymbol{\phi}}{\operatorname{argmin}} \left[\sum_i -(1-y_i) \log\Big[1 - \operatorname{sig}[\mathrm{f}[\mathbf{x}_i, \boldsymbol{\phi}]]\Big] - y_i \log\Big[\operatorname{sig}[\mathrm{f}[\mathbf{x}_i, \boldsymbol{\phi}]]\Big] \right]$$

Discriminator: generated samples, y = 0, real examples, y = 1:

$$\hat{\boldsymbol{\phi}} = \underset{\boldsymbol{\phi}}{\operatorname{argmin}} \left[\sum_j -\log\Big[1 - \operatorname{sig}[\mathrm{f}[\mathbf{x}^*_j, \boldsymbol{\phi}]]\Big] - \sum_i \log\Big[\operatorname{sig}[\mathrm{f}[\mathbf{x}_i, \boldsymbol{\phi}]]\Big] \right]$$

GAN cost function

Discriminator uses standard cross entropy loss:

$$\hat{\boldsymbol{\phi}} = \underset{\boldsymbol{\phi}}{\text{argmin}} \left[\sum_i -(1-y_i) \log\Big[1 - \text{sig}[\text{f}[\mathbf{x}_i, \boldsymbol{\phi}]]\Big] - y_i \log\Big[\text{sig}[\text{f}[\mathbf{x}_i, \boldsymbol{\phi}]]\Big] \right]$$

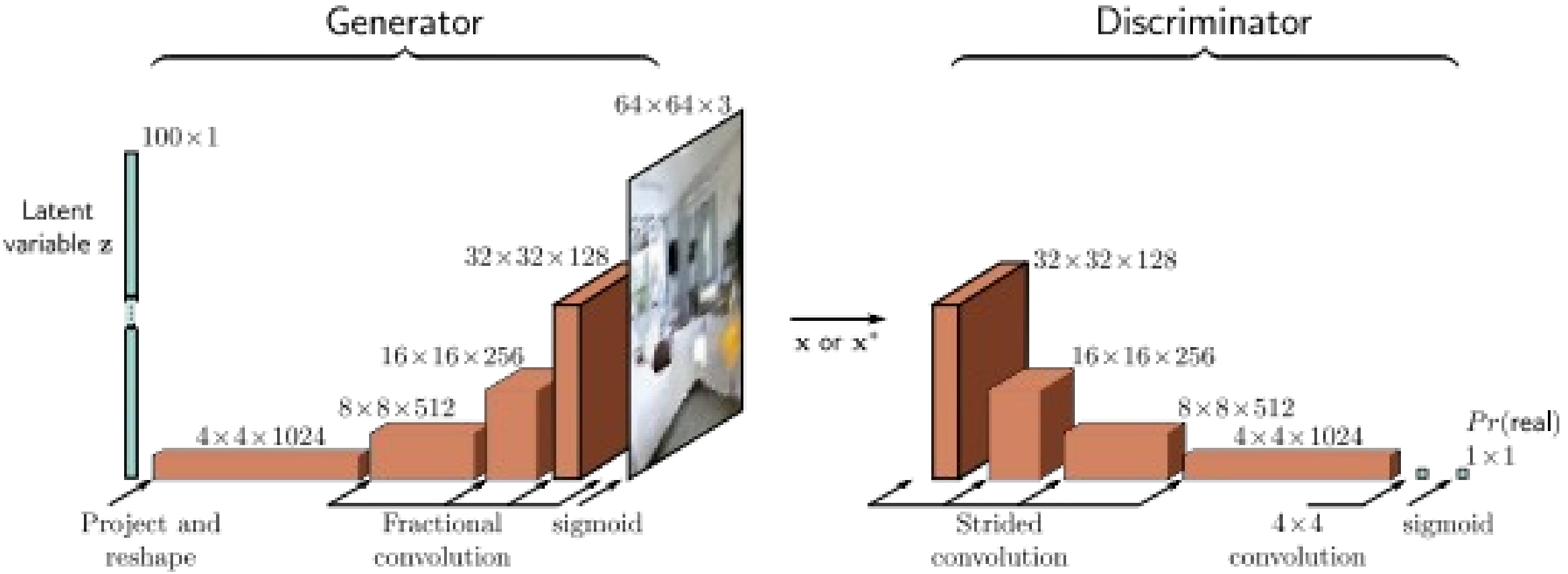
Discriminator: generated samples, y = 0, real examples, y = 1:

$$\hat{\boldsymbol{\phi}} = \underset{\boldsymbol{\phi}}{\text{argmin}} \left[\sum_j - \log\Big[1 - \text{sig}[\text{f}[\mathbf{x}^*_j, \boldsymbol{\phi}]]\Big] - \sum_i \log\Big[\text{sig}[\text{f}[\mathbf{x}_i, \boldsymbol{\phi}]]\Big] \right]$$

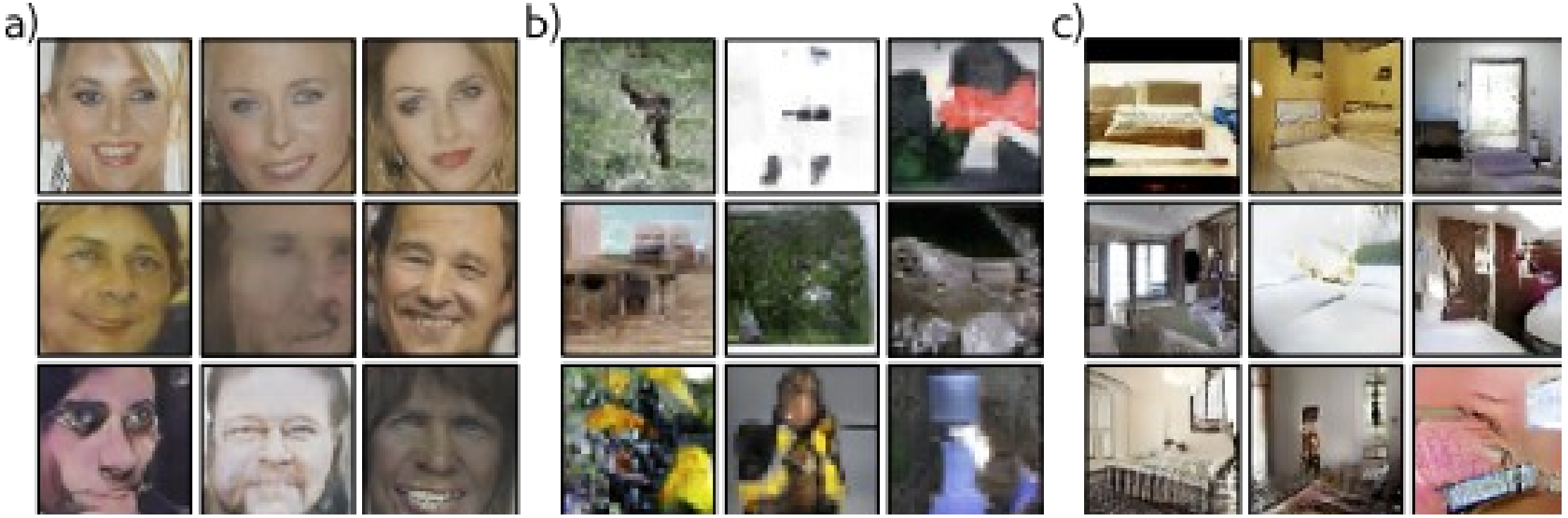
Generator loss: make generated samples more likely under discriminator (i.e. make discriminator loss larger)

$$\hat{\boldsymbol{\phi}}, \hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\phi}}{\text{argmin}} \left[\underset{\boldsymbol{\theta}}{\text{argmax}} \left[\sum_j - \log\Big[1 - \text{sig}[\text{f}[\mathbf{g}[\mathbf{z}_j, \boldsymbol{\theta}], \boldsymbol{\phi}]]\Big] - \sum_i \log\Big[\text{sig}[\text{f}[\mathbf{x}_i, \boldsymbol{\phi}]]\Big] \right] \right]$$

Deep Convolutional (DC) GAN

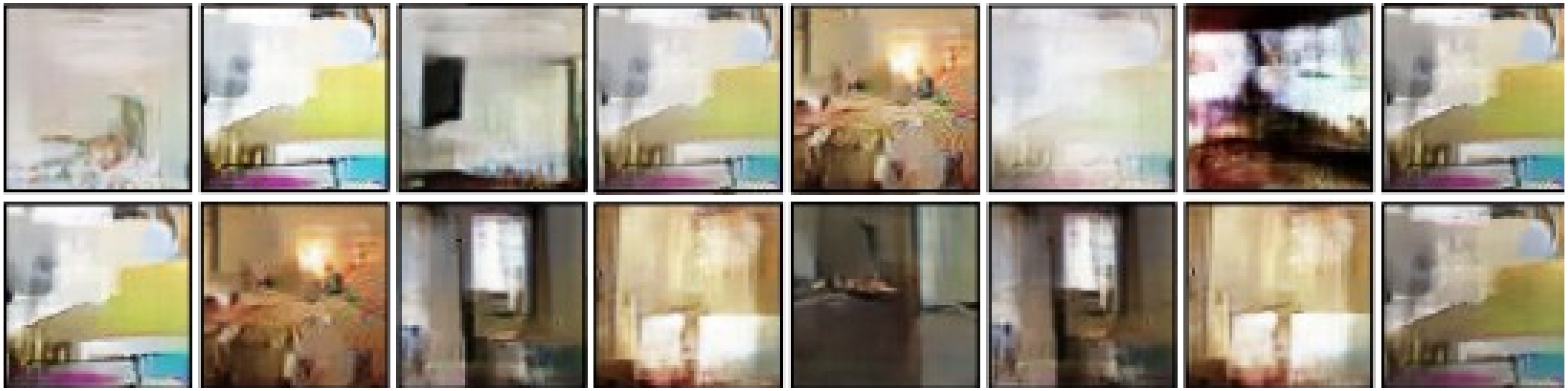


DC GAN Results

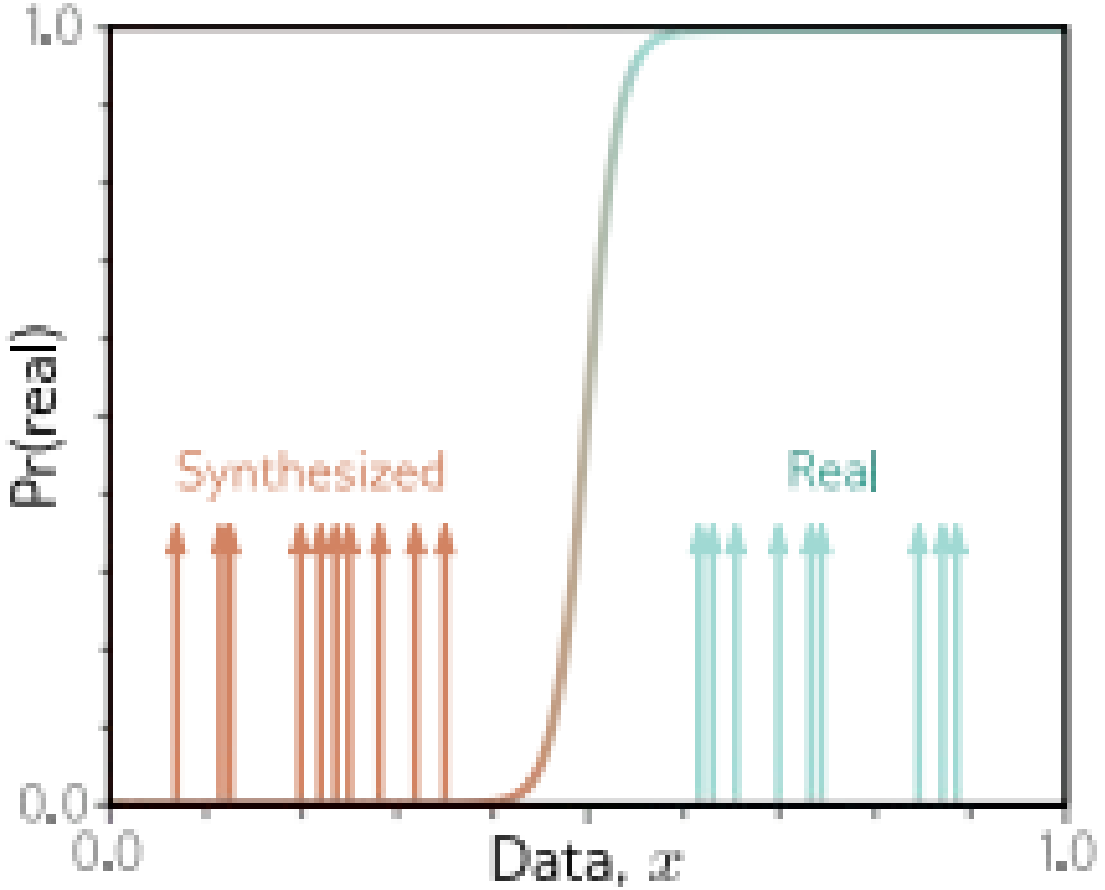


“Mode collapse”

Training GANs is difficult. Often fails.



GAN training instability



Solve with Wasserstein GAN loss function?

- Original function:

$$\hat{\phi} = \underset{\phi}{\text{argmin}} \left[\sum_j -\log\left[1 - \text{sig}[\text{f}[\mathbf{x}_j^*, \phi]]\right] - \sum_i \log\left[\text{sig}[\text{f}[\mathbf{x}_i, \phi]]\right] \right]$$

- Wasserstein GAN:

$$L[\phi] = \sum_j \text{f}[\mathbf{x}_j^*, \phi] - \sum_i \text{f}[\mathbf{x}_i, \phi]$$

subject to:

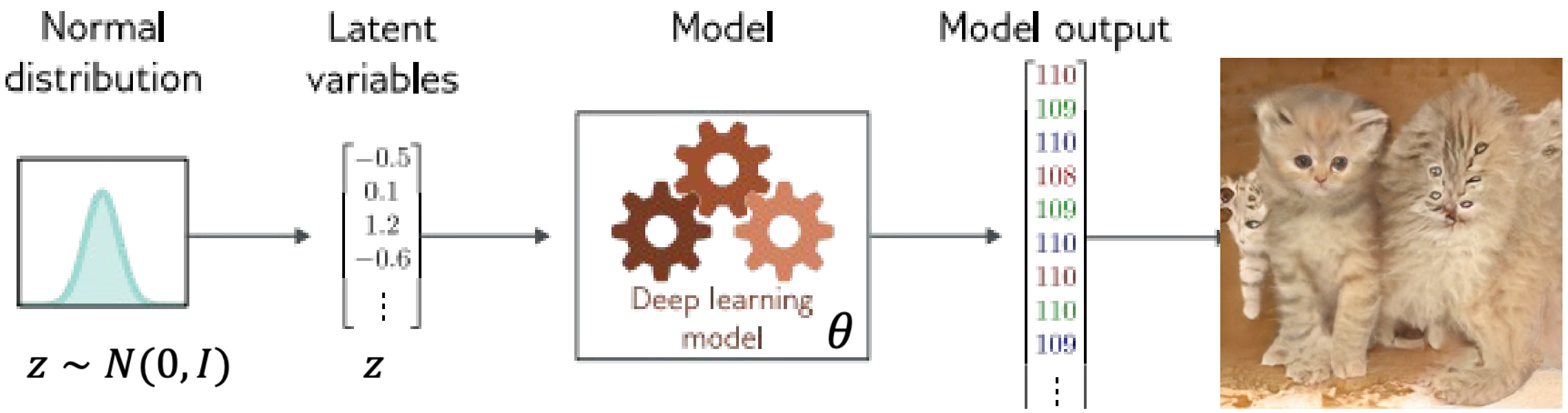
$$\left| \frac{\partial \text{f}[\mathbf{x}, \phi]}{\partial \mathbf{x}} \right| < 1$$

Example results

More generative model tricks

Most of these first illustrated with GANs, but now also done using other methods

Latent variable models (VAE or GAN)



Latent variable models map a random “latent” variable to create a new data sample
In GANs, this is called the generator:

$$\mathbf{x}_* = \mathbf{g}[\mathbf{z}_i, \boldsymbol{\theta}]$$

Interpolation (in latent space)

A common follow-up: Can we design so that certain dimensions control certain qualities?

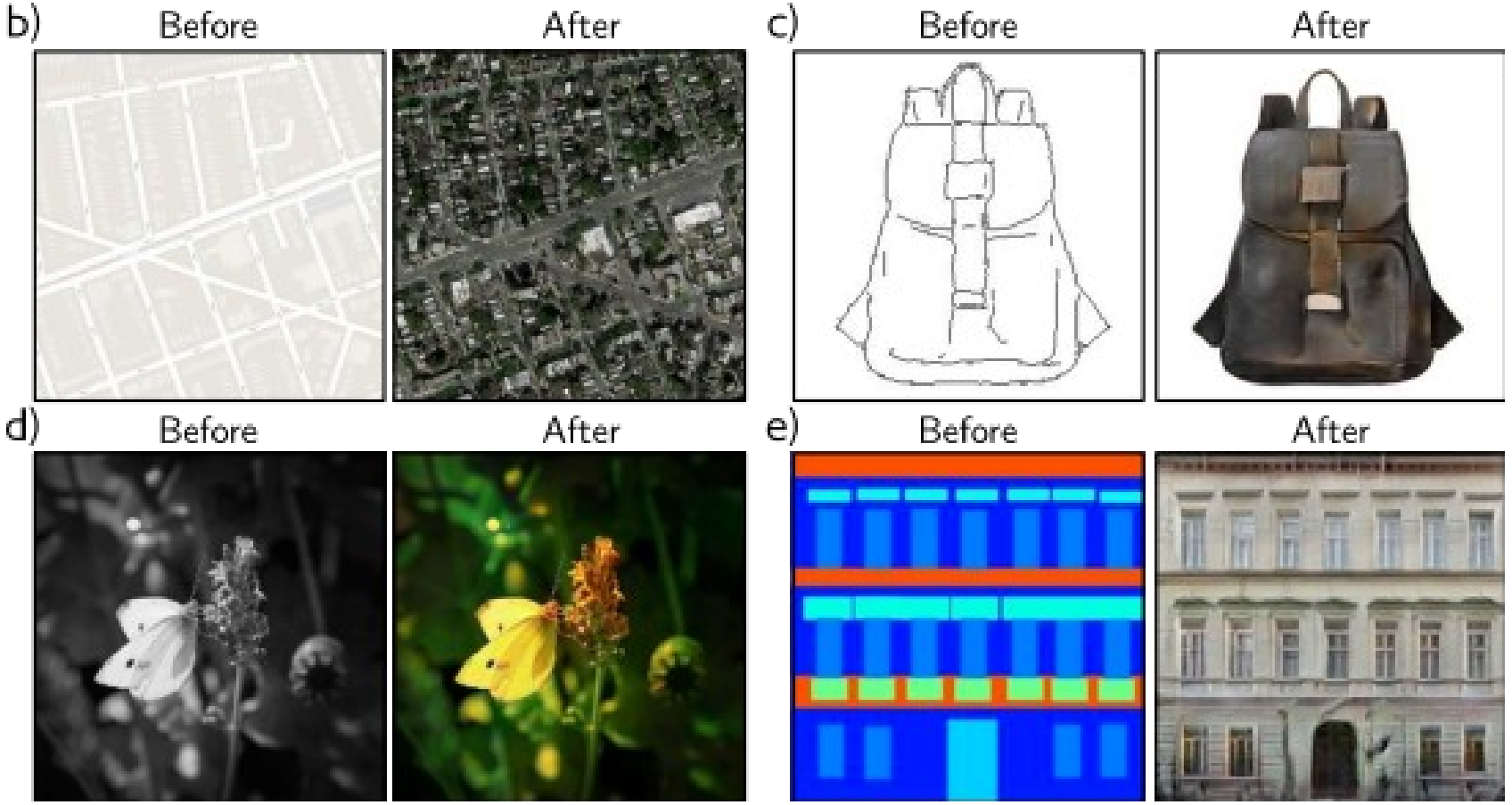
Image Translation

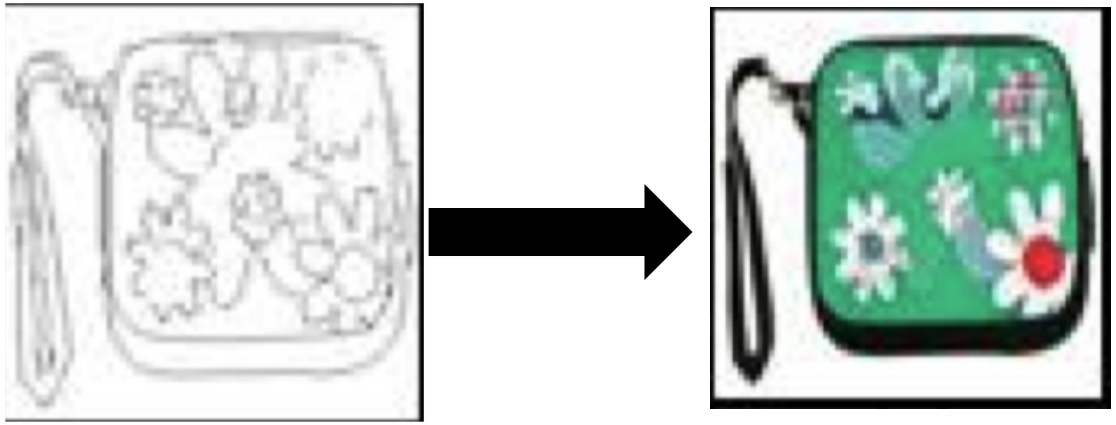
Methods:

- ControlNet + diffusion
- Diffusion bridges
- GANs
- Regular supervised learning
- ?

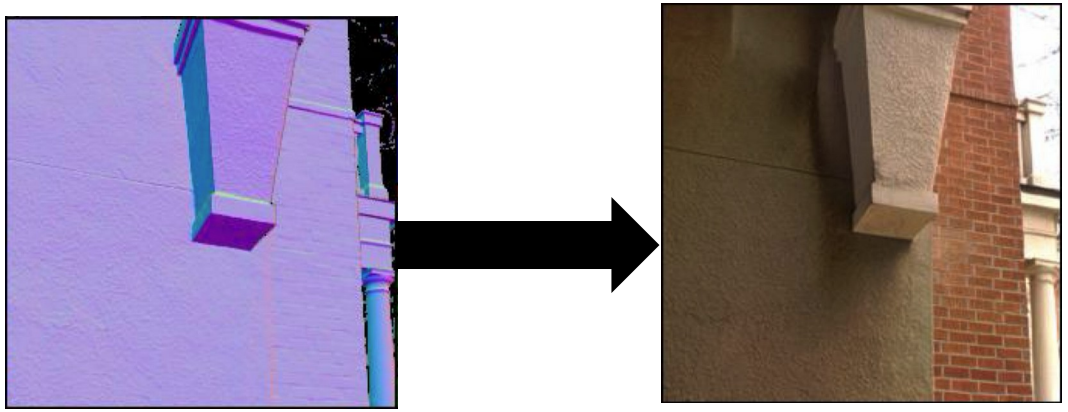
Image translation: Pix2Pix

"paired image-to-image translation"

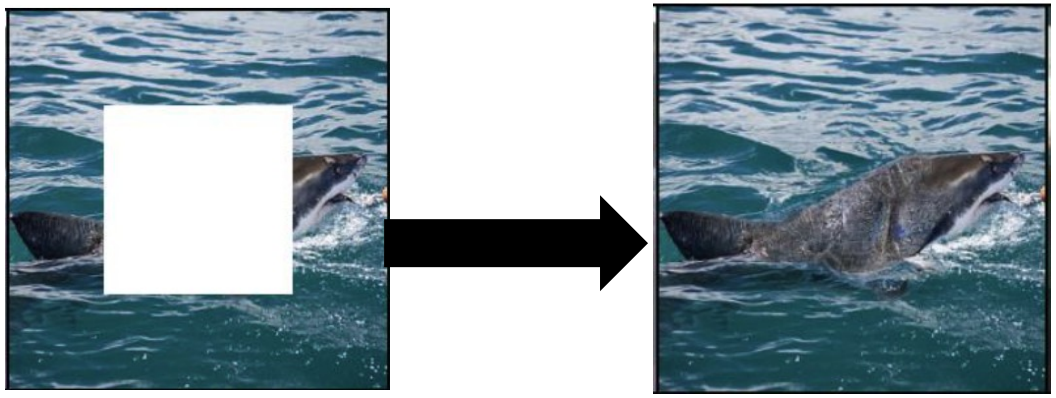




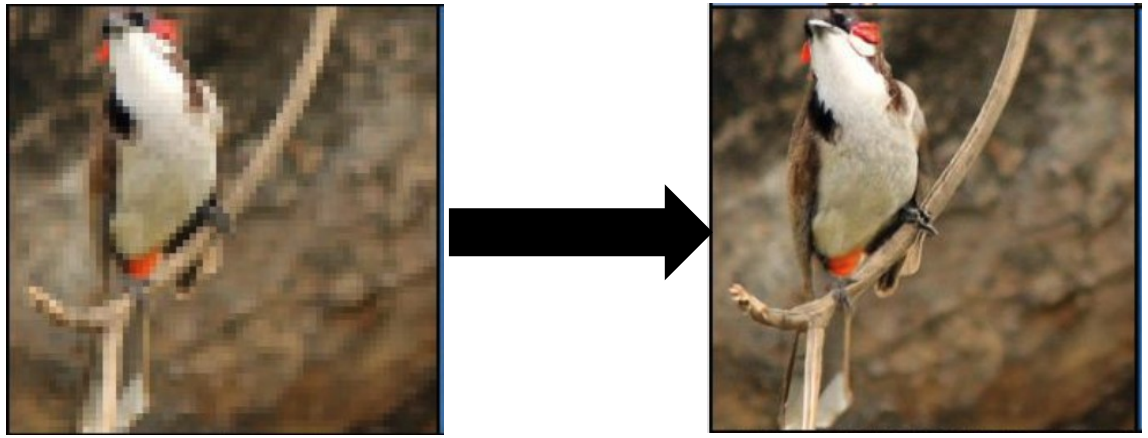
Edges to handbags



Depth to RGB images



Inpainting



Super-resolution

Image translation GAN style: Pix2Pix

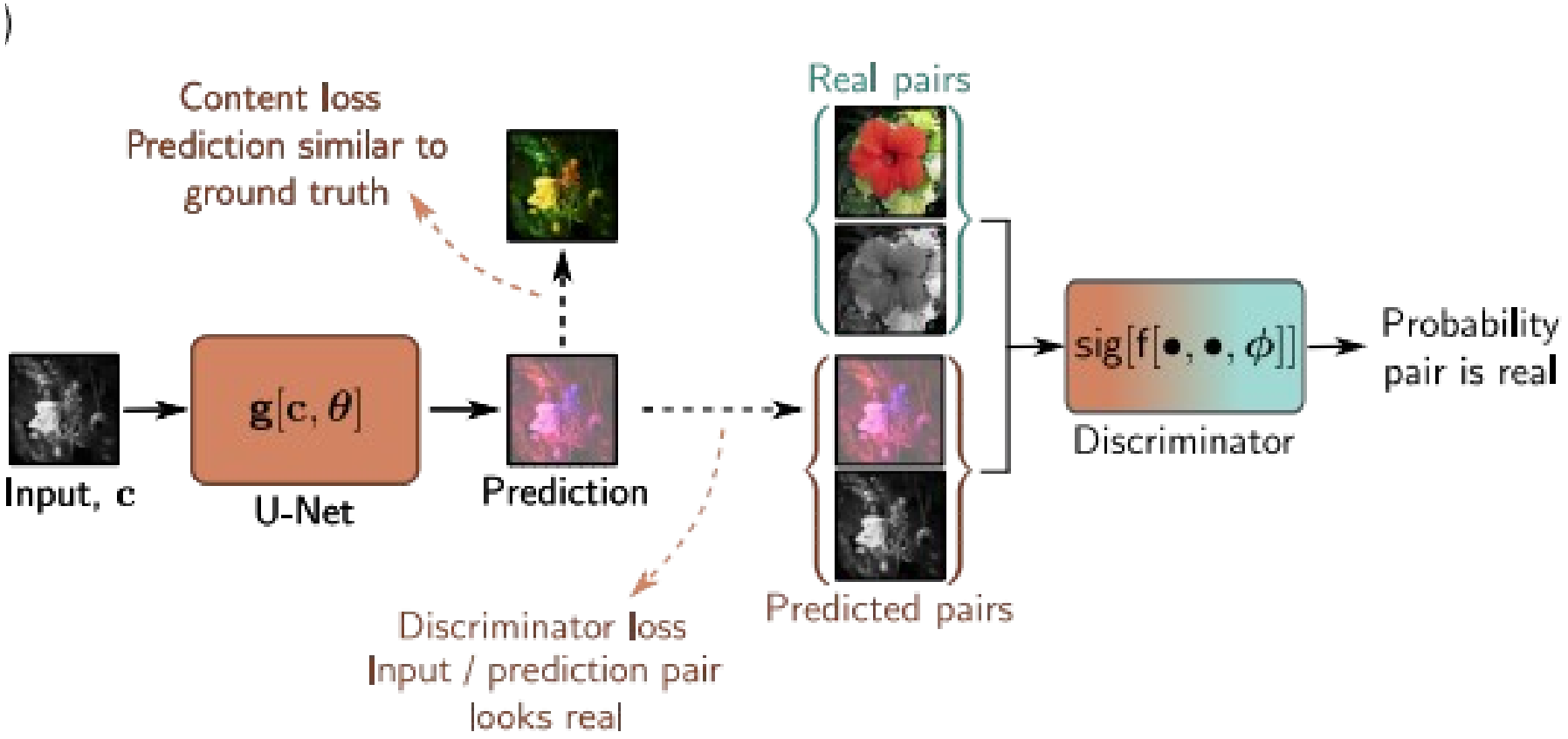


Image translation example: Super-resolution

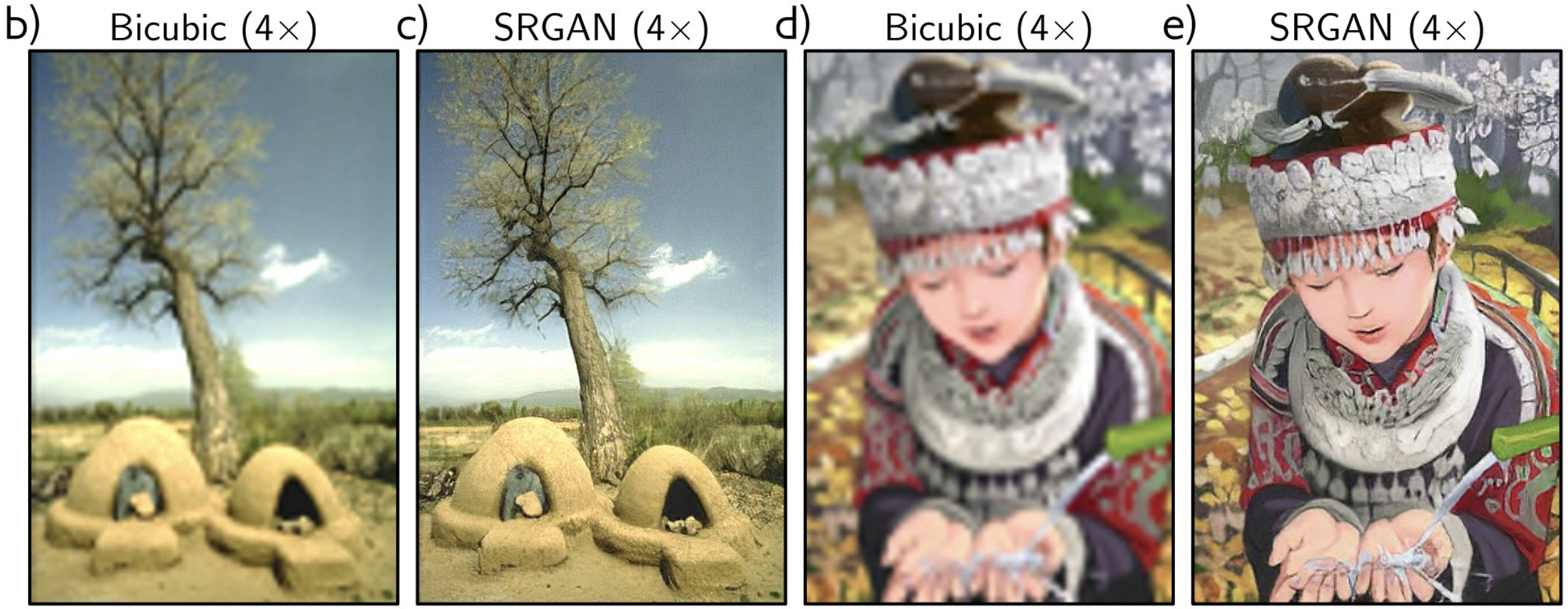
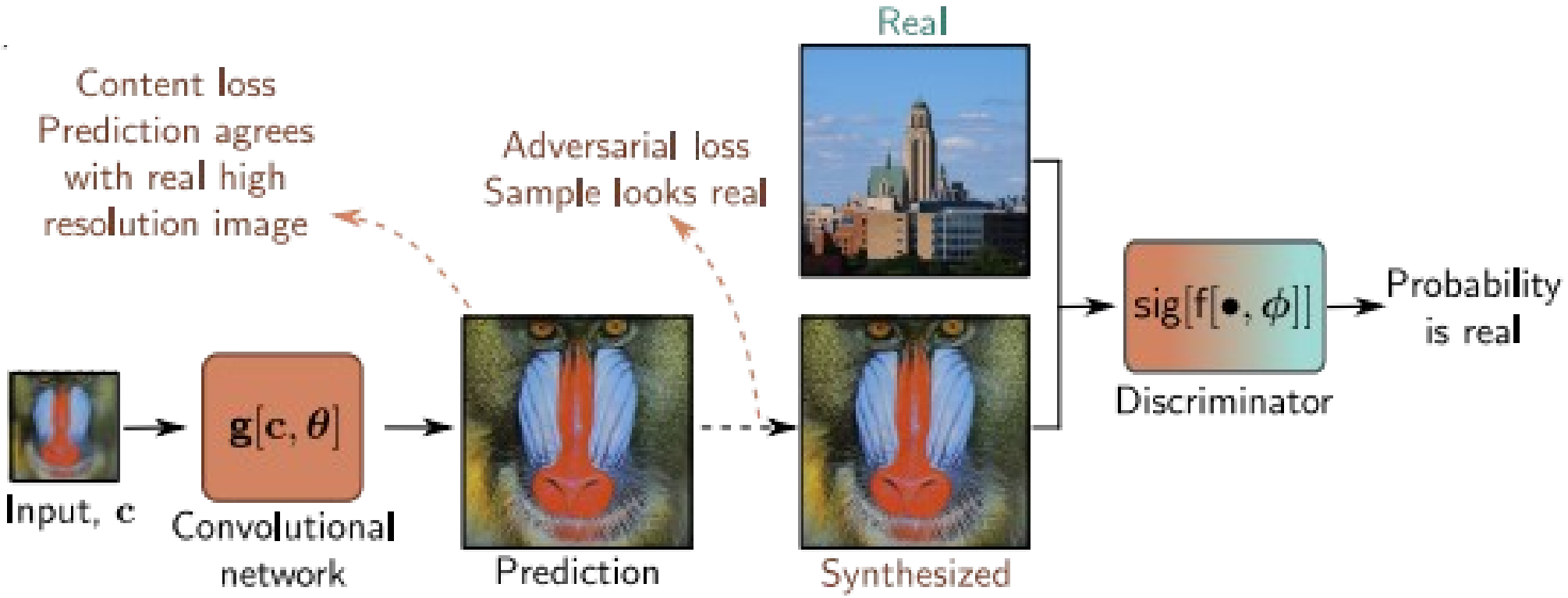
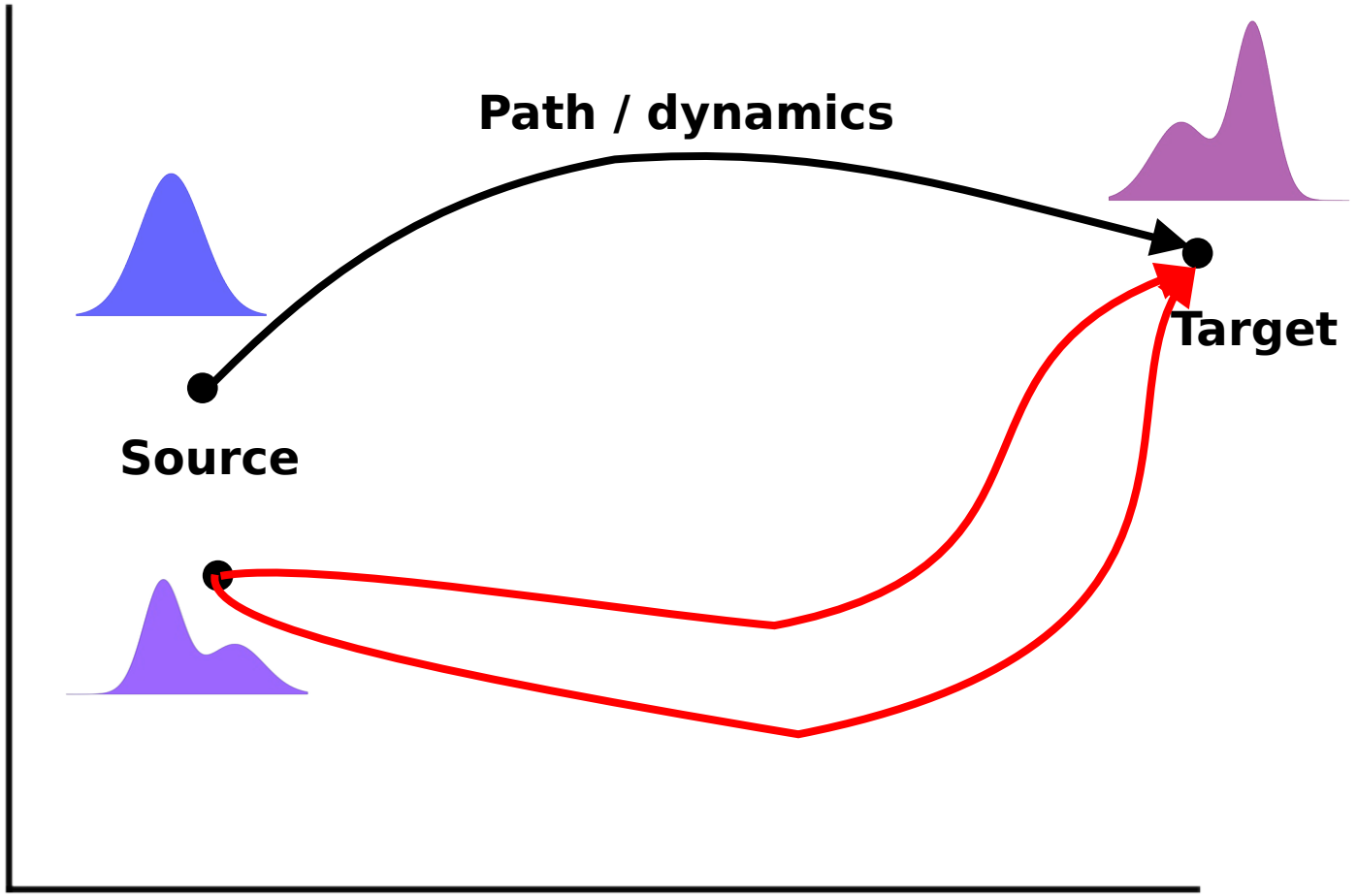


Image translation: Super Resolution GAN



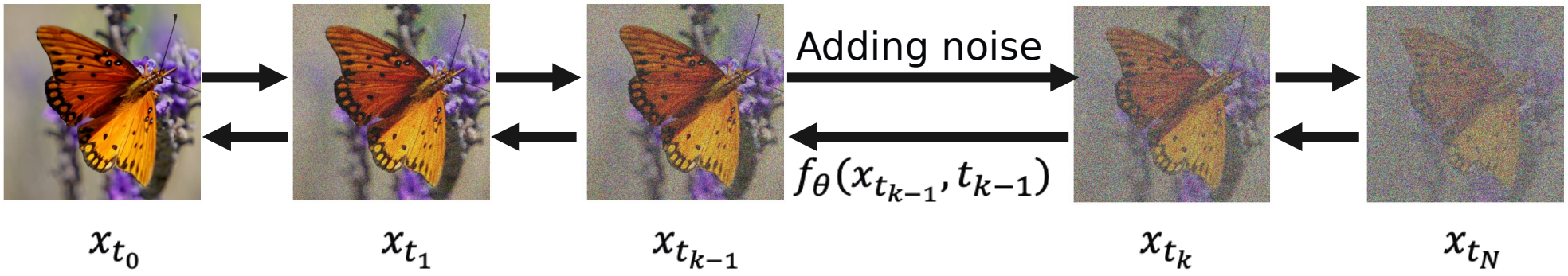
Diffusion bridges



DDBM. Zhou L, Lou A, Khanna S, et al. Denoising Diffusion Bridge Models[C]//The Twelfth International Conference on Learning Representations.
DBIM. Zheng K, He G, Chen J, et al. Diffusion bridge implicit models[J]. arXiv preprint arXiv:2405.15885, 2024.
DSBM. Shi Y, De Bortoli V, Campbell A, et al. Diffusion schrödinger bridge matching[J]. Advances in Neural Information Processing Systems, 2023, 36: 62183-62223.
SI. Albergo M S, Boffi N M, Vanden-Eijnden E. Stochastic interpolants: A unifying framework for flows and diffusions[J]. arXiv preprint arXiv:2303.08797, 2023.
Shaorong Zhang, .. Ver Steeg group: NeurIPS 2025, Exploring the Design Space of Diffusion Bridge Models via Stochasticity Control

Diffusion versus Diffusion Bridges

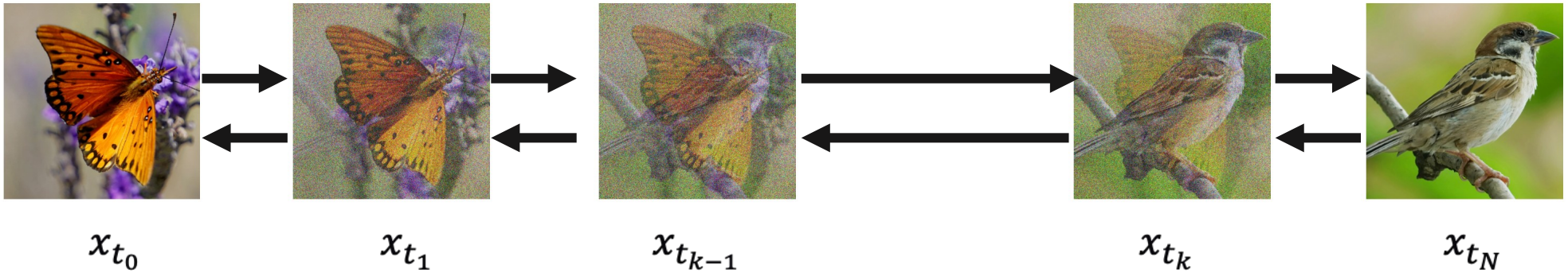
nsport between Gaussian and target distribution



Diffusio n model

A special case

ansport between any two distributions



Diffusion bridge model

Stochastic control for better *conditional diversity* in sampling

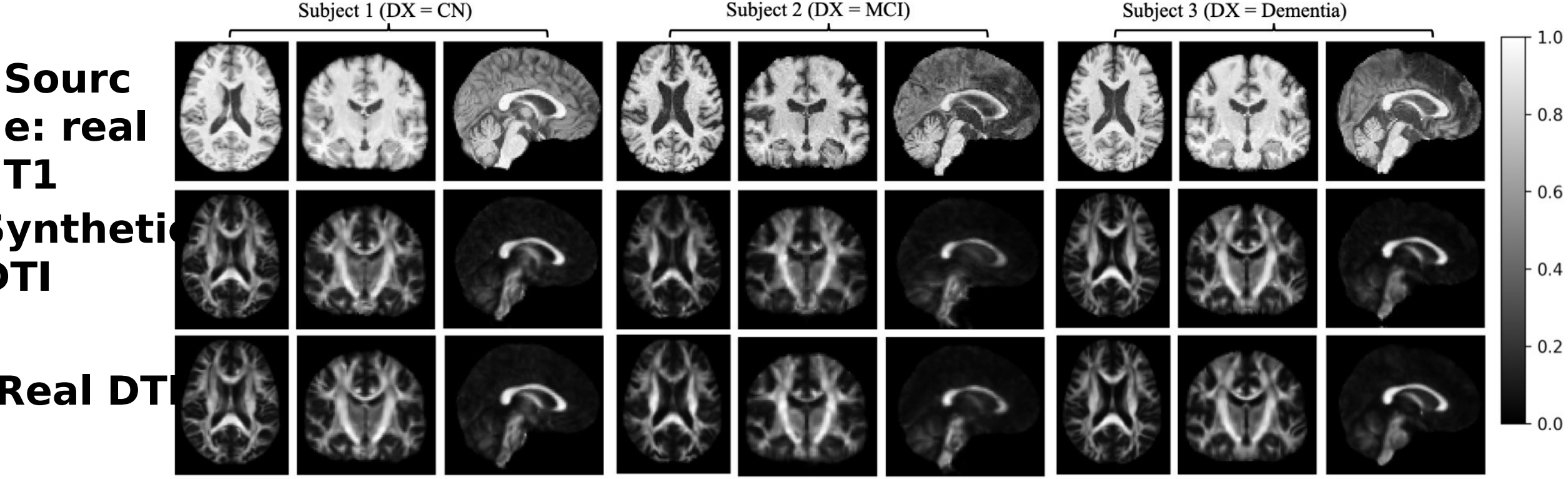
Ours

DDBM

Measuring conditional diversity

Model	Sampler	NFEs ↓	FID ↓	AFD ↑
DDBM-VP	DDBM	118	1.83	6.99
DDBM-VP	SDB	40	1.16	6.29
Ours	SDB	**5**	**0.89**	6.00
	SDB	40	1.71	**9.52**

Medical Image Translation



- Visually, very good recovery of DTI
- Downstream task performance (Alzheimer’s classification) almost as good! (90%, synthesized VS 91%, real)

“unpaired image-to-image translation”

Pictures of horses

(Unrelated) Pictures of zeb

Without changing *any other details of the image*, change this horse to a zebra

Image translation: CycleGAN

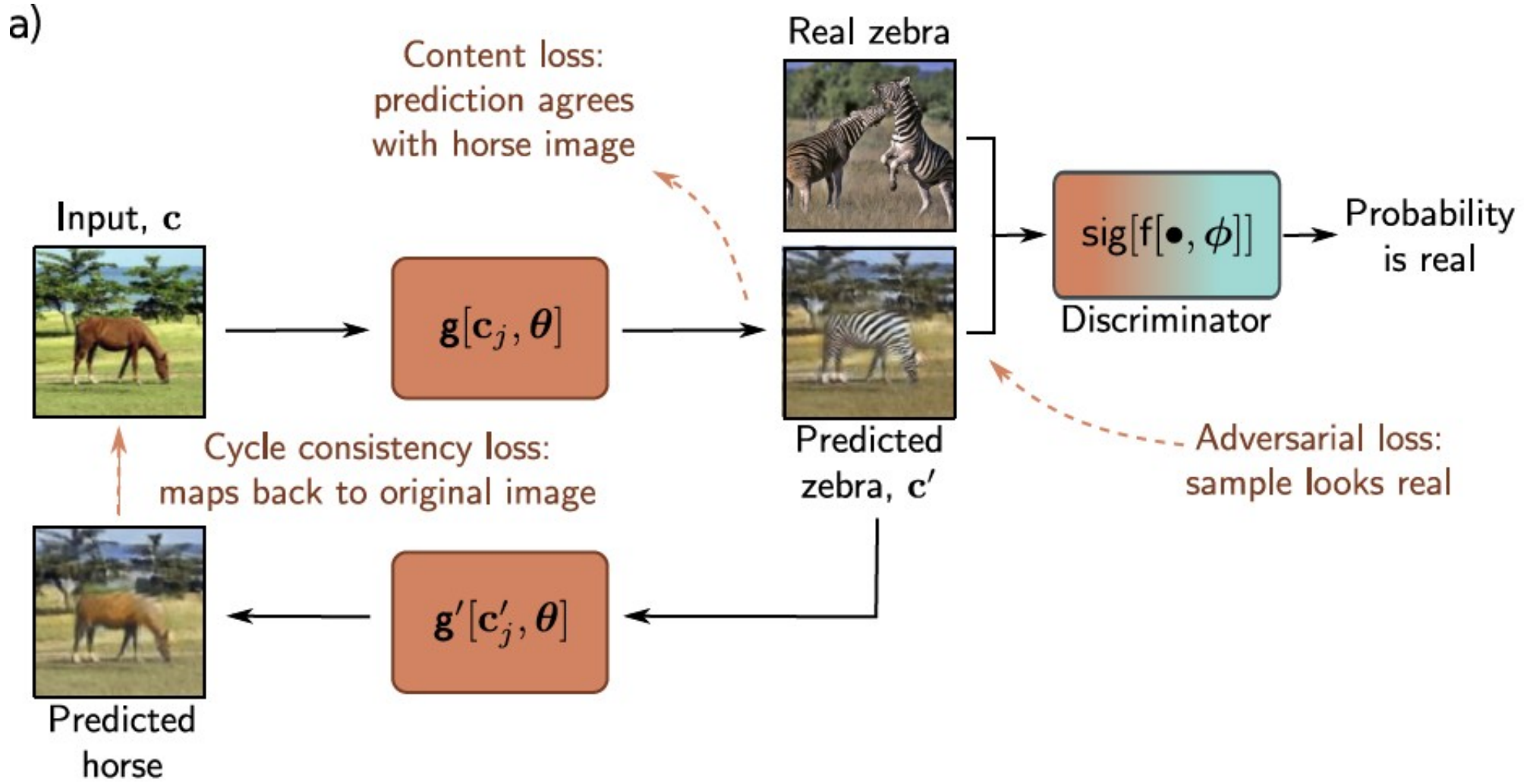
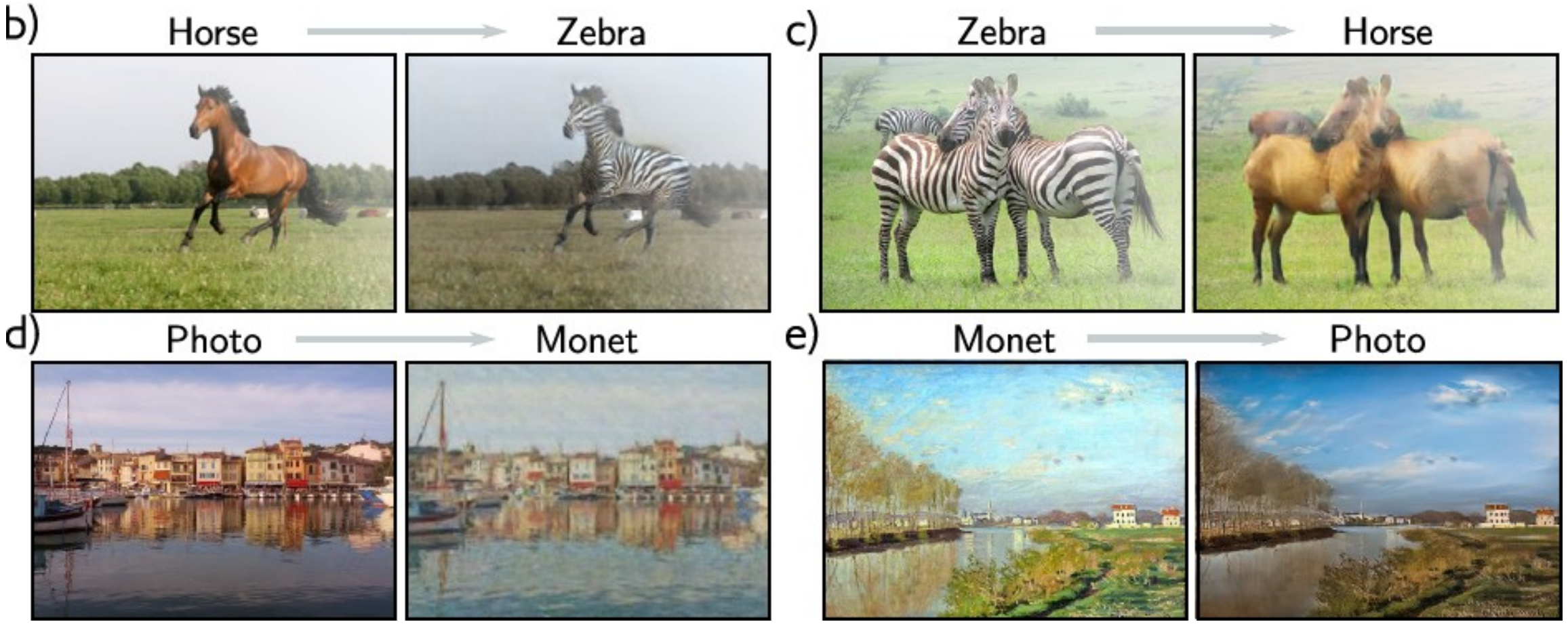


Image translation: CycleGAN



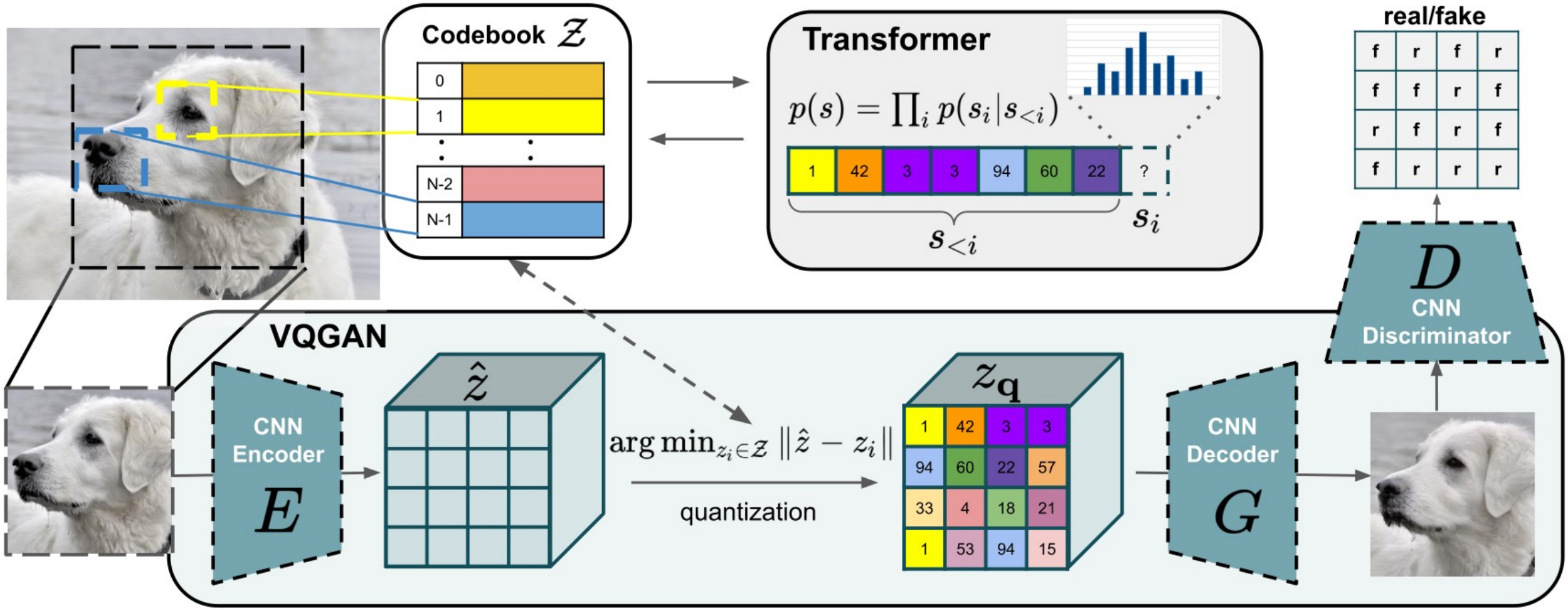
Example limitations from original

https://github.com/junyanz/CycleGAN

WEDNESDAY:
Variational Auto-Encoder (VAE)
And VQ-VAE, VQ-GAN

Vector Quantized: VQ-GAN

https://compvis.github.io/taming-transformers


Codebook $\mathcal{Z}$
0
1
N-2
N-1
Transformer
$p(s) = \prod_i p(s_i|s_{<i})$
1 42 3 3 94 60 22 ?
$s_{<i}$
s_i
real/fake
f r f r
f f r f
r f r f
f r r r
D
CNN Discriminator
VQGAN
CNN Encoder
E
$\hat{z}$
$\arg\min_{z_i \in \mathcal{Z}} \|\hat{z} - z_i\|$
quantization
$z_{\mathbf{q}}$
1 42 3 3
94 60 22 57
33 4 18 21
1 53 94 15
CNN Decoder
G